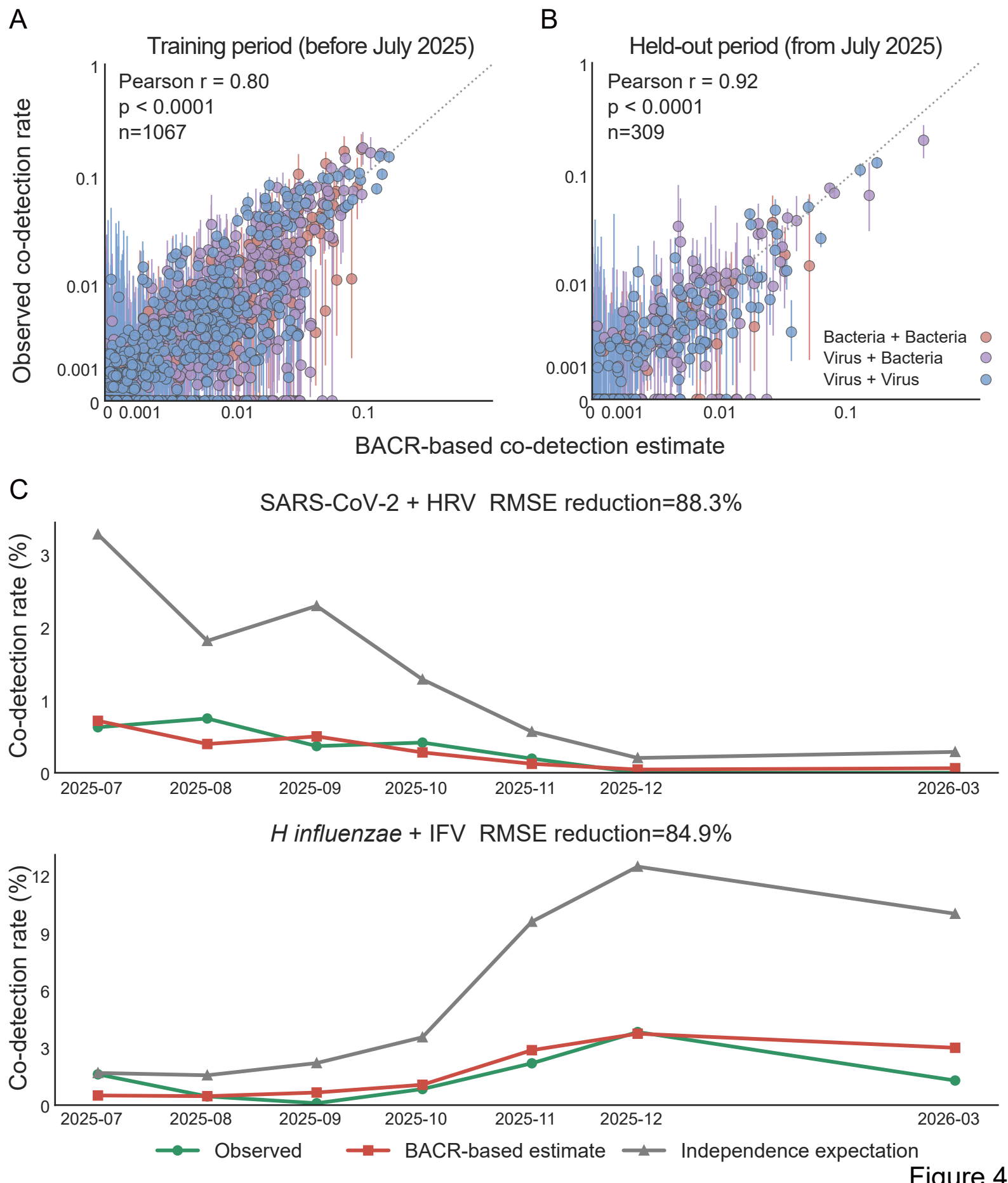

A
Training period (before July 2025)
Pearson r = 0.80
p < 0.0001
n=1067
Observed co-detection rate
B
Held-out period (from July 2025)
Pearson r = 0.92
p < 0.0001
n=309
Bacteria + Bacteria
Virus + Bacteria
Virus + Virus
BACR-based co-detection estimate
C
SARS-CoV-2 + HRV RMSE reduction=88.3%
Co-detection rate (%)
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-03
H influenzae + IFV RMSE reduction=84.9%
Observed
BACR-based estimate
Independence expectation

Figure 4